Problem Statement:

1. Image to Raw text extraction (OCR)
2. Image to Structured ONDC Product Object (JSON)
3. Structured Product Input to Structured ONDC Product Object (JSON)

1. Raw text extraction from Image (OCR)

1. Approaches :
    1. Vision Language Model
    2. Layout Detection per region + Text Extraction - VLM
2. Models tested :
    1. Chandra OCR 2 :
        1. Size: 4B
        2. Quantization: 4 bit
        3. Time:
        4. Handles Indic Languages
    2. Gemma 4 :
        1. Size: E4B/ 12B
        2. Quantization: 4 bit
        3. Time:
        4. E4B performs better than 12B
        5. Draft Models available for faster inference
    3. PaddleOCR-VLM :
        1. Size: 0.9B
        2. Quantization: None (bf16)
        3. Time:
        4. Requires Layout Detection for good performance
    4. Qwen 3.5 :
        1. Size: 4B/ 9B
        2. Quantization: 4 bit
        3. Time:
3. Evaluation. :
    - Character Accuracy = 1 - Character Error Rate
    -
4. Results:
    - ONCD Dataset

Model	Backend	Mean Acc	Avg Time
Qwen 3.5 9B	Mlx VLM	67.0%	34.6s
Gemma 4 E4B	Mlx VLM	66.5%	1.7s
Qwen 3.5 4B	Mlx VLM	65.2%	3.9s
Gemma 4 12B	Mlx VLM	55.9%	11.9s
PaddleOCR	PP-DocLayoutV3 + VLM	52.2%	1.7s
Chandra OCR	Mlx VLM	38.9%	11.5s

2. Image to Structured ONDC Product Object (JSON)

1. Approaches :
    1. Image → Structured Output
    2. Image → OCR → Structured Output
    3. Structured Output Generation:
        1. Prompt :
            1. Simpler approach
            2. No Output Degradation
            3. Due to LM's non deterministic nature — output structure not guaranteed
        2. Constrained Generation:
            1. Block illegal tokens & follow strict schema
            2. Output structure 100% Guaranteed
            3. May cause Output degradation
            4. Tools:
                1. Outlines - universal backend agnostic library, but currently does not support large VLMs
                2. LLGuidance - supported for all lm as well as vlm for mlx backend
2. Models tested:
    1. Gemma 4:
        1. Size : E4B / 12 B
        2. Quantization : 4 bit
        3. Time : 6.8s / 36.2s
        4. Draft models available
    2. Qwen 3.5:
        1. Size : 4B / 9B
        2. Quantization : 4 bit

        3. Time : 7.2s / 15.4s
3. Evaluation:
    1. Precision : Percentage of how many fields extracted are correct
    2. Recall : Percentage of how many fields extracted are present in our product structure
    3. F1 : Harmonic mean of Precision & Recall — Used [0-1]
4. Results:

Model	Pipeline	Precision	Recall	F1	Avg Time
Qwen 3.5 9B	Image -> JSON	72.1	71.2	70.2	15.4
Gemma 4 E4B	OCR -> JSON	68.2	71.2	68.6	6.8
Qwen 3.5 4B	OCR -> JSON	74.8	66.7	66.5	7.2
Gemma 4 12B	Image -> JSON	69.6	67.4	65.0	36.2

3. Structured Product Input to Structured ONDC Product Object (JSON)

1. Approaches :
    1. Structured Product Input → Structured Product Output
        1. Prompt :
            1. Simpler approach
            2. No Output Degradation
            3. Due to LM's non deterministic nature — output structure not guaranteed
        2. Constrained Generation:
            1. Block illegal tokens & follow strict schema
            2. Output structure 100% Guaranteed
            3. May cause Output degradation
            4. Tools:
                1. Outlines - universal backend agnostic library, but currently does not support large VLMs
                2. LLGuidance - supported for all lm as well as vlm for mlx backend
2. Models tested:
    1. Gemma 4:
        1. Size : E4B / 12 B
        2. Quantization : 4 bit
        3. Time :
        4. Draft models available
    2. Qwen 3.5:

        1. Size : 4B / 9B
        2. Quantization : 4 bit
        3. Time :
3. Evaluation:
    1. Precision : Percentage of how many fields extracted are correct
    2. Recall : Percentage of how many fields extracted are present in our product structure
    3. F1 : Harmonic mean of Precision & Recall — Used [0-1]
4. Results:
    - ONDC Dataset

Model	Precision	Recall	F1	Acc	Avg Time
Gemma 4 E4B	74.5	76.6	75.4	70	
Qwen 3.5 4B	33.0	79.7	43.1	30.7	

- ONDC Dataset:
    1. Grocery product details
    2. 5644 samples with 41 product features
    3. each sample has 3-5 images of the product
- ONDC Product Taxonomy:
    4. class ProductExtraction():
        1. Category
        2. Sub_Category
        3. Product_Name
        4. Net_Quantity
        5. UOM
        6. Bullet_Point
        7. Manufacturer
        8. Country_Of_Origin
        9. Images
        10. Brand
        11. Pack_Quantity
        12. Pack_Size
        13. UPC_or_EAN
        14. FSSAI_no
        15. Dimension
        16. Preservatives
        17. Preservatives_Details
        18. Flavours_or_Spices
        19. Diet_Type
        20. Ready_to_cook

21. Ready_to_eat
22. Ingredients
23. Nutritional_Details
24. Usage_Details
25. Storage_Instructions
26. Recommended_Age
27. Baby_Weight
28. Absorption_Duration
29. Scented_Flavoured
30. Herbal_or_Ayurvedic
31. Theme_or_Occasion_Type
32. Hair_Type
33. Mineral_Source
34. Caffeine_Content
35. Capacity
36. Variants